

fn x_mul_2k

Michael Shoemate (@Shoeboxam)

This proof resides in “**contrib**” because it has not completed the vetting process.

Proves soundness of the implementation of `x_mul_2k` in `mod.rs` at commit `f5bb719` (outdated¹).

1 Hoare Triple

Precondition

Compiler-verified

None

Caller-verified

$k \neq \text{i32.MIN}$

Pseudocode

```
1 def x_mul_2k(x: RBig, k: i32) -> RBig:
2     num, den = x.into_parts() #
3     if k < 0:
4         den <= cast(-k, usize)
5     else:
6         num <= cast(k, usize)
7
8     return RBig.from_parts(num, den)
```

Postcondition

Theorem 1.1. Return $x \cdot 2^k$.

Proof. Since x is a fraction, line 2 splits x into its numerator and denominator.

Consider two cases:

- If $k < 0$, then multiplying the denominator by 2^{-k} is equivalent to multiplying by 2^k . Since k is negative, then $-k$ is positive, meaning the cast to `usize` is valid. Shifting $-k$ zeros to the left is equivalent to multiplying the denominator by 2^k . Negation of k is well-defined for all values of `i32`, except for `i32.MIN`, which is not allowed by the precondition. Therefore, the result is $x \cdot 2^k$.
- If $k \geq 0$, then we multiply the numerator by 2^k . Since k is positive, the cast to `usize` is valid. Shifting k zeros to the left is equivalent to multiplying the numerator by 2^k . Therefore, the result is $x \cdot 2^k$.

In both cases, the result is $x \cdot 2^k$. □

¹See new changes with `git diff f5bb719..c09efd0 rust/src/measurements/noise/nature/float/utilities/mod.rs`